Dataset overview

Data	No of rows	No of columns	No. of LDs	No. of FDs
Global house purchase dataset.csv	10000	23	1658	166

Logical dependencies (LDs)

LDs capture rule-based relationships that hold within specific subsets of the data

Tilda (~>) denotes LDs, q indicates the strength of the dependency (lower is stronger)

Example context city = Abu Dhabi | salary < 10000

	Determinant (LHS)		Dependent (RHS)	strength (q)
Layer 1 (single attribute)	{ city }	~>	{ monthly_expenses }	q= 0.0616
	{ constructed_year }	~>	{ price }	q= 0.0156
	{ property_size_sqft }	~>	{ satisifaction_score }	q= 0.1084
Layer 2 (two attributes)	{ crime_case_reported, legal_case }	~>	{ decision }	q= 0.1765
	{ city, loan_tenure_years }	~>	{ emi_to_income_ration }	q= 0.1583
	{ property_size_sqft, rooms }	~>	{ decision }	q= 0.0420
Layer 3 (three attributes)	{ city, constructed_year, bathrooms }	~>	{ decision }	q= 0.1344
	{ city, constructed_year, neighbourhood_rating }	~>	{ legal_cases_on_property }	q= 0.0855

Functional dependencies (FDs)

FDs capture deterministic relationships that hold globally across the entire data

Arrow denotes FDs

Layer 1 (single attribute)	{ city }	->	{ country }
Layer 2 (two attributes)	{ down_payment, neighbourhood_rating }	->	{ decision }

Dataset overview

Data	No of rows	No of columns	No. of LDs	No. of FDs
Global student digital behaviour.csv	500	47	3519	396

Logical dependencies (LDs)

LDs capture rule-based relationships that hold within specific subsets of the data

Tilda (~>) denotes LDs, q indicates the strength of the dependency (lower is stronger)

Example context field of study = Business | stress level < 8

	Determinant (LHS)		Dependent (RHS)	strength (q)
Layer 1 (single attribute)	{ academic_risk_score }	~>	{ cyberbullying_exposure }	q= 0.1250
	{ field_of_study }	~>	{ stress_level }	q= 0.1653
	{ sleep_hours }	~>	{ late_night_usage }	q= 0.1250
Layer 2 (two attributes)	{ age, field_of_study }	~>	{ country }	q= 0.1548
	{ education_level, cyberbullying_exposure }	~>	{ productivity_score }	q= 0.1429
	{ development_level, adult_content_ exposure }	~>	{ academic_motivation }	q= 0.1653
Layer 3 (three attributes)	{ average_internet_speed, age, cyberbullying_exposure }	~>	{ adult_content_exposure }	q= 0.1641
	{ development_level, education _level, late_night_usage }	~>	{ poverty_rate_percent }	q= 0.1303

Functional dependencies (FDs)

FDs capture deterministic relationships that hold globally across the entire data

Arrow denotes FDs

Layer 1 (single attribute)	{ class_attendance_rate }	->	{ academic_risk_score }
	{ late_night_score }	->	{ late_night_usage }
Layer 2 (two attributes)	{ age, anxiety_score }	->	{ late_night_usage }
	{ age, anxiety_score }	->	{ late_night_score }